MCQ from All Topics (According to Slides)

Greedy Technique

Q1. What is the main idea of a greedy algorithm?

A) Divide problem into subproblems
B) Choose the best option at each step
C) Explore all possibilities
D) Use recursion only

Answer: B) Choose the best option at each step

Q2. Greedy algorithms are mainly used for —

A) Sorting problems
B) Searching problems
C) Optimization problems
D) Traversal problems

Answer: C) Optimization problems

Q3. Which property is necessary for greedy algorithms?

A) Divide and Conquer
B) Dynamic Allocation
C) Greedy Choice Property
D) Binary Property

Answer: C) Greedy Choice Property

Q4. Which problem may fail using greedy approach?

A) Fractional Knapsack
B) Coin Change
C) Job Scheduling
D) MST

Answer: B) Coin Change

Fractional Knapsack

Q5. In Fractional Knapsack, items can be —

A) Taken only once
B) Divided into fractions
C) Ignored completely
D) Taken twice

Answer: B) Divided into fractions

Q6. Which method gives the best result in Fractional Knapsack?

A) Smaller weight first
B) Larger weight first
C) Profit/Weight ratio
D) Random selection

Answer: C) Profit/Weight ratio

Minimum Spanning Tree

Q7. A spanning tree must contain —

A) All edges
B) All vertices
C) Cycles
D) Directed edges

Answer: B) All vertices

Q8. Minimum Spanning Tree contains —

A) Maximum cost
B) Minimum cost
C) Maximum vertices
D) Duplicate edges

Answer: B) Minimum cost

Q9. Which algorithm is used for MST?

A) Dijkstra
B) Bellman-Ford
C) Kruskal
D) DFS

Answer: C) Kruskal

Q10. An MST of a graph with n vertices contains —

A) n edges
B) n+1 edges
C) n−1 edges
D) n^2 edges

Answer: C) n−1 edges

Kruskal's Algorithm

Q11. Kruskal's Algorithm is based on —

A) Dynamic Programming
B) Greedy Method
C) Divide and Conquer
D) BFS

Answer: B) Greedy Method

Q12. Kruskal's Algorithm selects —

A) Maximum edge first
B) Random edge first
C) Minimum edge first
D) Negative edge first

Answer: C) Minimum edge first

Q13. What happens if an edge forms a cycle in Kruskal’s Algorithm?

A) Added
B) Multiplied
C) Rejected
D) Reversed

Answer: C) Rejected

Q14. Which data structure is commonly used in Kruskal’s Algorithm?

A) Stack
B) Queue
C) Priority Queue
D) Array

Answer: C) Priority Queue

Single Source Shortest Path

Q15. SSSP stands for —

A) Single Source Smallest Path
B) Single Source Shortest Path
C) Smallest Source Shortest Path
D) Simple Source Shortest Path

Answer: B) Single Source Shortest Path

Q16. In weighted graphs, path cost is calculated using —

A) Number of vertices
B) Number of paths
C) Sum of edge weights
D) Multiplication of nodes

Answer: C) Sum of edge weights

Dijkstra’s Algorithm

Q17. Dijkstra’s Algorithm is used for —

A) Sorting
B) MST
C) Shortest Path
D) Graph Coloring

Answer: C) Shortest Path

Q18. Dijkstra’s Algorithm follows —

A) BFS
B) DFS
C) Greedy Approach
D) Divide and Conquer

Answer: C) Greedy Approach

Q19. Dijkstra’s Algorithm does not work with —

A) Positive edges
B) Weighted graphs
C) Negative edges
D) Directed graphs

Answer: C) Negative edges

Q20. Initial distance of source node in Dijkstra is —

A) ∞
B) 1
C) -1
D) 0

Answer: D) 0

Q21. Initial distance of other nodes in Dijkstra is —

A) 0
B) 1
C) ∞
D) −∞

Answer: C) ∞

Q22. What is the process of updating shortest distance called?

A) Traversal
B) Relaxation
C) Recursion
D) Rotation

Answer: B) Relaxation

Bellman-Ford Algorithm

Q23. Bellman-Ford Algorithm works with —

A) Only positive edges
B) Negative edge weights
C) Unweighted graphs
D) Cyclic trees only

Answer: B) Negative edge weights

Q24. Bellman-Ford Algorithm relaxes edges —

A) Once
B) Twice
C) $V-1$ times
D) V^2 times

Answer: C) $V-1$ times

Q25. Bellman-Ford detects —

A) MST
B) Topological order
C) Negative weight cycle
D) Hamiltonian path

Answer: C) Negative weight cycle

Q26. A change in the V-th iteration indicates —

A) BFS completion
B) MST formation
C) Negative cycle
D) Graph coloring

Answer: C) Negative cycle

Topological Sort

Q27. Topological sorting is possible only for —

A) Undirected Graph
B) Cyclic Graph
C) DAG
D) Weighted Graph

Answer: C) DAG

Q28. DAG stands for —

A) Directed Acyclic Graph
B) Double Acyclic Graph
C) Directed Adjacent Graph
D) Dynamic Acyclic Graph

Answer: A) Directed Acyclic Graph

Q29. Topological sorting represents —

A) Weight ordering
B) Dependency ordering
C) Color ordering
D) Tree ordering

Answer: B) Dependency ordering

Q30. Which graph cannot have topological ordering?

A) DAG
B) Directed Graph
C) Cyclic Graph
D) Tree

Answer: C) Cyclic Graph

Kahn’s Algorithm

Q31. Kahn’s Algorithm is based on —

A) DFS
B) BFS
C) DP
D) Greedy

Answer: B) BFS

Q32. Kahn’s Algorithm starts with vertices having indegree —

A) 1
B) −1
C) 0
D) 2

Answer: C) 0

Q33. Indegree means —

A) Number of outgoing edges
B) Number of incoming edges
C) Number of cycles
D) Number of paths

Answer: B) Number of incoming edges

DFS Based Topological Sort

Q34. Which data structure is used in DFS topological sort?

A) Queue
B) Stack
C) Heap
D) Tree

Answer: B) Stack

Q35. In DFS topological sort, nodes are pushed into stack —

A) Before DFS
B) During initialization
C) After visiting neighbors
D) Randomly

Answer: C) After visiting neighbors

Backtracking

Q36. Backtracking builds solution —

A) Randomly
B) Incrementally
C) Iteratively only
D) Without recursion

Answer: B) Incrementally

Q37. What happens when backtracking reaches an invalid state?

A) Program stops
B) Restart occurs
C) Algorithm backtracks
D) Graph resets

Answer: C) Algorithm backtracks

Q38. Which is an application of backtracking?

A) N Queens
B) MST
C) Dijkstra
D) Bellman-Ford

Answer: A) N Queens

Graph Coloring

Q39. Graph coloring ensures adjacent vertices have —

A) Same color
B) Different colors
C) Same weight
D) Same degree

Answer: B) Different colors

Q40. The minimum number of colors required for a graph is called —

A) Vertex Number
B) Degree Number
C) Chromatic Number
D) Color Factor

Answer: C) Chromatic Number

Q41. Coloring a graph with at most m colors is called —

A) m-pathing
B) m-covering
C) m-coloring
D) m-graphing

Answer: C) m-coloring

N Queens Problem

Q42. In N Queens Problem, queens must not attack each other on —

A) Rows only
B) Columns only
C) Diagonals only
D) Rows, columns, and diagonals

Answer: D) Rows, columns, and diagonals

Q43. Which technique is used in N Queens Problem?

A) Greedy
B) Backtracking
C) BFS
D) DP

Answer: B) Backtracking

Branch and Bound

Q44. Branch and Bound is mainly used for —

A) Searching
B) Optimization
C) Traversal
D) Sorting

Answer: B) Optimization

Q45. In Branch and Bound, branching means —

A) Removing nodes
B) Dividing into subproblems
C) Sorting edges
D) Coloring nodes

Answer: B) Dividing into subproblems

Q46. Pruning means —

A) Expanding nodes
B) Removing unnecessary branches
C) Creating cycles
D) Adding edges

Answer: B) Removing unnecessary branches

Q47. FIFO Branch and Bound uses —

A) Stack
B) Queue
C) Heap
D) Array

Answer: B) Queue

Approximation Algorithm

Q48. Approximation algorithms are mainly used for —

A) Easy problems
B) NP-hard problems
C) Sorting problems
D) Tree traversal

Answer: B) NP-hard problems

Q49. Approximation algorithms provide —

A) Exact solution always
B) Near-optimal solution
C) No solution
D) Random output

Answer: B) Near-optimal solution

Traveling Salesman Problem

Q50. TSP stands for —

A) Total Shortest Path
B) Traveling Salesman Problem
C) Tree Spanning Problem
D) Traveling Source Path

Answer: B) Traveling Salesman Problem

Q51. TSP requires visiting each city —

A) Twice
B) Randomly
C) Exactly once
D) Infinitely

Answer: C) Exactly once

Q52. TSP is a —

A) Polynomial problem
B) NP-hard problem
C) Sorting problem
D) DFS problem

Answer: B) NP-hard problem

Q53. TSP approximation algorithm uses —

A) DFS Tree
B) Binary Tree
C) Minimum Spanning Tree
D) Heap Tree

Answer: C) Minimum Spanning Tree

Q54. Triangle Inequality states that direct path should be —

A) More expensive
B) Less or equal expensive
C) Infinite
D) Negative

Answer: B) Less or equal expensive

Vertex Cover Problem

Q55. A vertex cover covers —

A) Vertices only
B) Cycles only
C) All edges
D) Only source node

Answer: C) All edges

Q56. Vertex Cover Problem is generally defined on —

A) Directed Graph
B) Undirected Graph
C) Tree only
D) DAG only

Answer: B) Undirected Graph

Q57. Goal of Vertex Cover Problem is to find —

A) Maximum spanning tree
B) Minimum vertex set
C) Shortest path
D) Longest edge

Answer: B) Minimum vertex set

Q58. A real-life application of Vertex Cover is —

A) CPU scheduling
B) Traffic camera placement
C) File compression
D) Encryption

Answer: B) Traffic camera placement